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OPTIMAL SCREENING PLANS FOR NONSERIAL PRODUCTION SYSTEMS*†

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Quality control screening programs are defined and evaluated for the *general n-stage nonserial production process*. A total expected cost criterion, developed as a function of the screening applied at each inspection station, embraces the various costs of inspection, defect repair and defects passing through the process undetected. Absorbing Markov chains aid in the identification of probabilistic flows of defective materials through the production network.

Minimum cost screening programs are developed, giving the optimal level of screening at every potential inspection station. For a quasi-concave cost structure, it is shown that the optimal screening program will employ either zero or 100 percent effective screening throughout. A standard branch and bound "backtrack" strategy readily identifies optimal screening programs for the unconstrained (0, 1) nonlinear integer programming problem. An illustrative problem is presented.

1. Introduction

Business today, regardless of the nature of the product, is placing a greater emphasis on product quality. Competitive pressures have caused many firms to undertake product differentiation on quality alone. Increased costs of product maintenance and repair, together with increased reliability requirements, have generated both internal and external pressures upon firms to intensify their concern for product quality.

Many production processes—because of existing production techniques, equipment and raw materials—are technologically incapable of generating finished products of the desired quality. (Lindsay and Bishop [9] cite the manufacture of glass containers as an example.) Through the strategic location and use of inspection activities, screening programs can be developed to upgrade product quality and yield substantial economies of production.

There have been several recent contributions, [1], [3], [8], [9], [10], [11], and [12], developing optimal screening programs for serial (single-channel) production systems. Each identifies minimum-cost screening programs, balancing assumed linear inspection costs against per unit product repair or replacement costs and the costs of permitting defective product to reach the customer. Present day production processes are characterized more by nonserial networks in which input materials may take one of several paths through a production system. This paper investigates optimal screening programs for nonserial production processes operating under a nonlinear quasi-concave inspection cost structure.

The total cost of processing defective product includes such costs as the appraisal and detection of defects, internal and external failure costs.¹ In the absence of screening at the various stages of production, all defects eventually reach the customer.

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¹ This terminology has been adopted from Feigenbaum [4].

External failure costs result from the repair and replacement of this defective product according to contractual agreements, guarantees and goodwill policies. Through screening many defects can be detected and repaired prior to shipment to the customer. Internal failure costs, together with the costs of appraisal and detection of the screening program employed, are incurred. Optimal screening programs are those minimizing the cost of expected internal failures, expected external failures and the costs of appraisal and detection, hereafter called the total expected cost of quality.

In the following sections then (1) a model is formulated for the total expected cost of quality for a given screening program; and (2) a solution procedure is developed for the determination of optimal screening programs for nonserial production processes.

2. Outline of Problem and Solution

General Assumptions

Consider any nonserial production system in which raw materials enter the process through stage one and by subsequent production operations are transformed into N units of finished goods per unit time. Let stages $1, \dots, n-1$ represent the processing stages, with a finished goods warehouse n and one repair facility stage o . A unit of production is simply the resultant aggregate of materials and labor at the completion of any individual production stage. (For example, in the typical eight-stage system of Figure 1, both part 2 and subassembly A represent units of production.)

At every stage, materials and labor may be added to the incoming units of production. Let us call these exogenous materials and labor. Some of these may be defective. Hence, all defects are introduced into the system as: (1) defective raw material and labor through stage one and (2) defective exogenous material and labor through subsequent stages 2 through $n-1$.

Given a defective unit of exogenous material and labor added at stage i , the unit of

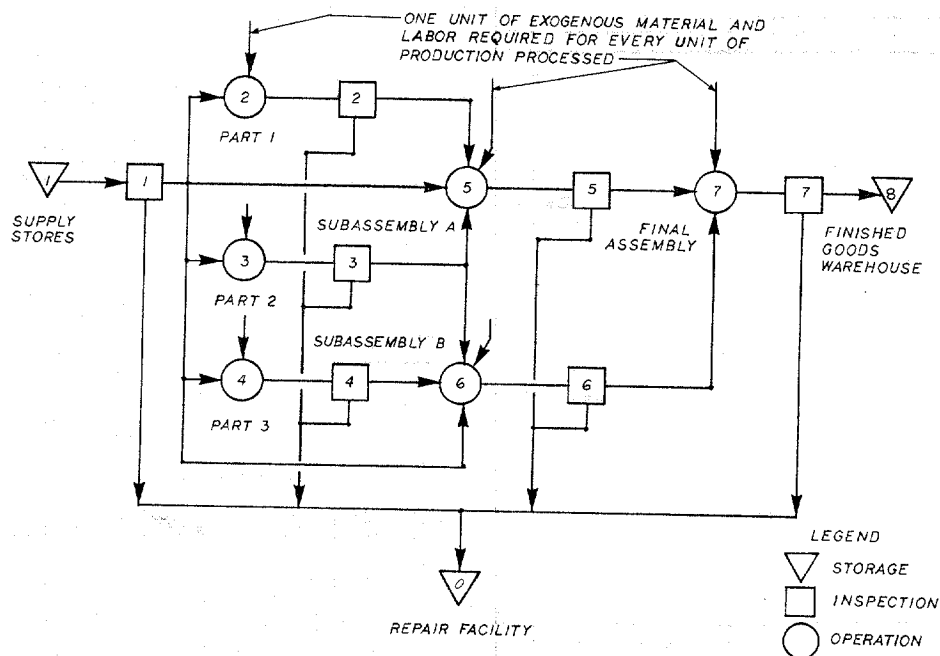


FIGURE 1. Typical eight-stage production process

production acquires a type i defect. A total of $n - 1$ defect types exist, where a type 1 defect denotes defective raw material. Clearly, a unit of production may contain several defect types.

Given that all defects may be removed and the unit of production repaired either in the field or in the repair facility, consider the following assumptions:

1. Associated with each production stage i is a potential inspection station where $0 \leq f_i \leq 1$ represents the decision variable screening effort or filtering effectiveness, and defines the probability of any defect being detected at that station. False detections are not considered.

2. All flows are assumed to be constant and under equilibrium conditions. Detected defects are retained in the repair facility as scrap. The repaired units of production are inserted back into the production process immediately following the inspection station at which the defects were discovered.

3. No inspection is undertaken at the repair facility o . Undetected defects entering the repair facility will remain undetected. They may, however, be discovered at subsequent inspection stations or in the field.

4. No inspection is undertaken at the finished goods warehouse. Once a defect reaches the warehouse, it is assumed to be shipped to the customer.²

Clearly the flow of defects may be described by transition probabilities. Eventually, all defects will be processed with their respective units of production and absorbed into the finished goods warehouse or the repair facility. The path to absorption depends upon several factors:

1. The design of the production process defining the expected flow rates between all stages of production;

2. The design of a quality control screening program outlining inspection rules and resulting levels of filtering effectiveness for every inspection station. Let this screening plan be given by the vector $F = (f_1, \dots, f_{n-1})$.

Expected Defect Flow Rates

Let column vector $X = (x_1, \dots, x_n)$ be the expected product flow vector in units of production with x_i the output from stage i required to support N units of finished goods per unit time. Let $Y = [y_{ij}]_{i,j=1}^n$ govern the distribution of flow within the process where $0 \leq y_{ij} \leq 1$ represents the fraction of the output from stage i directed to stage j . The flow is assumed to be circuitless; thus Y is a triangular matrix with $y_{ij} = 0$ for $i \geq j$, and since all units leaving stage i must transit to a stage of number $> i$, $\sum_{j=i+1}^n y_{ij} = 1$. Y is likewise assumed to be deterministic.³

² If screening is desired at warehouse n , a second hypothetical warehouse $n + 1$ in series with stage n can be inserted for which assumption four obtains.

³ (a) The vector X may be readily obtained from the relationship $X = (I - G)^{-1}\bar{N}$ where

I = an identity matrix;

$G = [g_{ij}]$ a connectivity matrix of 0 and 1 for $i, j = 1, \dots, n$;

$g_{ij} = 0$ if a direct connection does not exist;

$= 1$ if a direct connection does exist

from i to j .

$\bar{N} = (0, 0, \dots, N)$, N being the number of finished goods processed per unit time.

Similarly, the matrix Y may be obtained directly from G and X as follows: $y_{ij} = (x_j/x_i)g_{ij}$. Figure 3 gives an illustration of G .

(b) Since the flow is assumed circuitless, then $(I - G)$ may be written as a triangular matrix with unit elements along the diagonal. It can be shown that $|I - G| = 1 \neq 0$, and hence $(I - G)^{-1}$ exists.

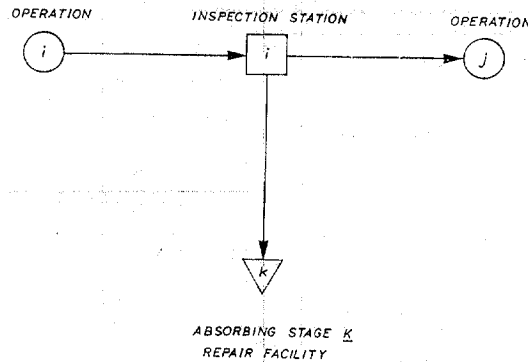


FIGURE 2. Elemental portion of a production system

Recall that defects may be introduced at any stage. Let $a = (a_1, \dots, a_{n-1})$ where a_i is the probability that a defect will be introduced into the process at stage i . In addition, let $A = [a_{ij}]_{i,j=1}^{n-1}$, where $a_{ij} = a_i$ if $i = j$ and $a_{ij} = 0$ otherwise.

In those production processes where $0 \leq y_{ij} \leq 1$, the movements of defects through the system can be described by simple absorbing Markov chains. Let stage $i = 1, \dots, n-1$ be transient states of the Markov process and stages $k = n, o$ be absorbing states. Consider first the elemental production system of Figure 2 and the evaluation of the transition probability matrix $P = [p_{ij}]_{i,j=1}^{n,o}$ of the Markov process.

The transition probability between any two nonabsorbing states i and j is simply $p_{ij} = (1 - f_i)y_{ij}$, $i, j = 1, \dots, n-1$. Similarly, the probabilities between any nonabsorbing state i and the absorbing states are $p_{ik} = f_i$ for $k = o$ the repair facility and by definition $p_{ik} = (1 - f_i)y_{ik}$ for $k = n$ the warehouse. By the definition of circuitless flow, $p_{ii} = 0$ for $i = 1, \dots, n-1$.

Let $B = [b_{ik}]_{i=1}^{n-1}$, $k = n, o$, where b_{ik} represents the probability of a defect leaving stage i and eventually progressing through the process to the input of stage $k = n, o$. The following recursive relationship holds $b_{ik} = p_{ik} + \sum_{j=1}^{n-1} p_{ij}b_{jk}$ for $1 \leq i \leq n-1$. It follows then that the absorbing probability matrix B is given by $B = (I - Q)^{-1}R$ where I is the identity matrix and $Q = [p_{ij}]_{i,j=1}^{n-1}$ and $R = [p_{ij}]_{i=1}^{n-1}$, $j = n, o$, are submatrices of the transition probability matrix P shown in canonical form below:⁴

$$P = \begin{bmatrix} I & 0 \\ R & Q \end{bmatrix}.$$

Let $P' = [p'_{ij}]_{i,j=1}^{n-1}$ where p'_{ij} represents the probability of a defect leaving stage i and eventually passing undetected to the input of stage j . The elements of P' must follow the recursive relationship $p'_{ij} = p_{ij} + \sum_{k=1}^{n-1} p_{ik}p'_{kj}$, or equivalently $P' = (I - Q)^{-1}Q$.

As a result, the expected number of type i defects absorbed into stage k per unit time is given by the expression $E = B^T A X$ where $E = [e_{ki}]_{i=1}^{n-1}$, $k = n, o$, $i = 1, \dots, n-1$, X is the appropriate $n-1$ components of the expected flow vector, and T denotes transposition.

Expected Cost of Quality Control

Assume that the total expected cost is of the form $T = C_D + C_E + C_I$ where C_D is the expected cost of appraisal and detection (cost of maintaining the screening

⁴ Given $P = \begin{bmatrix} I & 0 \\ R & Q \end{bmatrix}$, since $Q^n \rightarrow 0$ as $n \rightarrow \infty$, it can be shown that $(I - Q)^{-1}$ exists [6, p. 284].

program F), C_E is the expected external failure cost (costs of defects shipped to customers) and C_I is the expected internal failure cost (costs of internal repair of detected defects) of processing N units. Assume that these costs are given by:

$$C_D = \sum_{i=1}^{n-1} d_i$$

where $d_i \equiv d_i(f_i)$ is the cost of maintaining a screening program of f_i at station i with $d_i(0) = 0$. d_i is quasi-concave in the closed interval $(0, 1)$.⁵

$$C_E = \sum_{i=1}^{n-1} a_i x_i c_i^w b_{in}$$

where c_i^w is the per unit cost incurred as a defect introduced into stage i eventually passes undetected through to the finished goods warehouse facility n .

$$C_I = \sum_{i=1}^{n-1} a_i x_i (c_{ii}^r f_i + \sum_{j=1}^{n-1} c_{ij}^r p'_{ij} f_j)$$

where c_{ij}^r is the per unit cost incurred when a defect introduced at stage i is detected at inspection station j and sent to the repair facility.

Since $c_i^r b_{io} = c_{ii}^r f_i + \sum_{j=1}^{n-1} c_{ij}^r p'_{ij} f_j$, the internal failure cost can be re-written as follows:

$$C_I = \sum_{i=1}^{n-1} a_i x_i c_i^r b_{io}$$

where c_i^r is an averaged per unit cost of an internal defect introduced into stage i and repaired internally, absorbed into stage o .

Letting $C^w = (c_1^w, \dots, c_{n-1}^w)$, $C^r = (c_1^r, \dots, c_{n-1}^r)$ and $D = (d_1, \dots, d_{n-1})$, define the failure cost matrix

$$C = \begin{bmatrix} C^w \\ C^r \end{bmatrix}.$$

The total expected cost of quality T for given screening plan F becomes simply:

$$\begin{aligned} T &= \sum_{i=1}^{n-1} [d_i + a_i x_i (c_i^w b_{in} + c_i^r b_{io})] \\ &= S(D + (BC)^*AX) \\ &= S(D + ((I - Q)^{-1}RC)^*A(I - G)^{-1}\bar{N}) \end{aligned}$$

where S is an $(n - 1)$ component sum vector and U^* is the matrix resulting from setting all off-diagonal elements equal to zero, in any matrix U .

Results

THEOREM I. C_E and C_I are multi-linear forms in $f_1, \dots, f_{n-1}$.

PROOF. Let $M(f_i, \dots, f_j)$ designate any multi-linear form in the variables $f_i, \dots, f_j$. It is first shown that $b_{io} = M(f_i, \dots, f_{n-1})$ when $1 \leq i \leq n - 1$. Note that $b_{io} = p_{io} + \sum_{j=1}^{n-1} p_{ij} b_{jo}$. Assume that b_{jo} is of the indicated form for $j = i + 1, \dots, n - 1$ (this inductive hypothesis has been established for $j = n - 1$). Then, $b_{io} = f_i + \sum_{j=i+1}^{n-1} (1 - f_i) y_{ij} M(f_j, \dots, f_{n-1}) = M(f_i, \dots, f_{n-1})$ for all $1 \leq i \leq n - 1$. Thus, b_{io} has the prescribed form. For $k = n$, $b_{in} = p_{in} + \sum_{j=1}^{n-1} p_{ij} b_{jn} = (1 - f_i) y_{in} + \sum_{j=i+1}^{n-1} (1 - f_i) y_{ij} b_{jn}$. Similarly, it can be shown that b_{in} has the prescribed form.

By its definition, C_E and C_I will have the form claimed if $b_{i,k}$, $k = n, o$, does. This was just shown and the theorem is established.

Clearly, since C_D is by definition quasi-concave, then $T = C_D + C_E + C_I$ is also quasi-concave.

⁵ The cost of appraisal and detection is assumed to be a function of the decision variable f_i and may be linear or concave.

THEOREM II. Let σ be the unit hyper-cube in E^{n-1} and h . If for each $i = 1, \dots, n-1$, $h(f_1, \dots, f_{n-1})$ is concave with respect to f_i , given f_j fixed for all $j = 1, \dots, n-1$ ($j \neq i$), then h will take on every relative minimum value at an extreme point of the hyper-cube σ .

PROOF. For any $F = (f_1, \dots, f_{n-1})$, let $h_k(z/F) = h(f_1, \dots, f_{k-1}, z, f_{k+1}, \dots, f_{n-1})$. By hypothesis, $h_k(z/F)$ is concave in z for a fixed F and so it assumes its minimum over the closed interval $(0, 1)$ at either $z = 0$ or $z = 1$, that is

$$h_k(z/F) = \min [h_k(0/F), h_k(1/F)].$$

Let $F^0 = (f_1^0, \dots, f_{n-1}^0)$ be any point in σ and let $h(F^0)$ be the value of h at F^0 . Define $F^k = (f_1^k, \dots, f_{n-1}^k)$ recursively for $k = 1, \dots, n-1$ by:

$$\begin{aligned} f_i^k &= f_i^{k-1} \quad \text{for all } i \neq k, \\ &= 0 \quad \text{for } i = k \quad \text{and} \quad h_k(0/F^{k-1}) \leq h_k(1/F^{k-1}), \\ &= 1 \quad \text{for } i = k \quad \text{and} \quad h_k(0/F^{k-1}) > h_k(1/F^{k-1}). \end{aligned}$$

Then $h(F^k) \leq h(F^{k-1})$ for $k = 1, \dots, n-1$ and so $h(F^{n-1}) \leq h(F^0)$ for any F^0 (including a minimizing point). Since F^{n-1} is an extreme point solution, the theorem is established.

Solution Procedure

The problem reduces to one of minimizing T over all vectors $F = (f_1, \dots, f_{n-1})$ with 0, 1 components, and is an unconstrained integer (0, 1) programming problem with a nonlinear quasi-concave objective function. Partial enumeration using standard branch and bound methods and a "backtrack" strategy provide an efficient solution procedure given simple and sharp bounds.⁶

Let $T^* = \min T(F)$ and note that any feasible solution F provides an upper bound, $T(F)$, on T^* . An obvious and simple solution is $F^0 = (0, \dots, 0)$, the origin solution corresponding to no screening at all with an associated cost of $T(F^0) = \sum_{i=1}^{n-1} a_i x_i c_i^w$. Then, $T(F^0)$ provides an initial upper bound on T^* .

Consider any "partial solution" $F_m = (f_1, \dots, f_m)$ with $1 \leq m < n-1$. Let

$$U(F_m) = \sum_{i=1}^m [d_i(f_i) + a_i x_i (c_{ij}^r f_i + \sum_{k=i+1}^m c_{ik}^r p'_{ik} f_k)],$$

the detection cost for stages 1 through m plus the internal failure costs incurred as type i defects are introduced and detected within stages $i, \dots, m$. A lower bound on the failure costs of type j defects and the remaining undetected type i defects at stages j for $m < j \leq n-1$ is given by $L_m(j) = L_m(j)' + L_m(j)''$.

$$L_m(j)' = \min [\{d_j(1) + a_j x_j c_{jj}^r\}, \{\min_{j < k \leq n-1; p'_{ik} \neq 0} (a_j x_j p'_{jk} c_{jk}^r)\}, \{a_j x_j c_j^w\}],$$

the failure costs of type j defects detected internally at stage j , at some subsequent stage k , or passed on to the customer; and

$$\begin{aligned} L_m(j)'' &= \min [\{\sum_{i=1; p'_{ij} \neq 0}^m (a_i x_i p'_{ij} c_{ij}^r)\}, \{\sum_{i=1; p'_{ik} \neq 0}^m \min_{j < k \leq n-1} (a_i x_i p'_{ik} c_{ik}^r)\}, \\ &\quad \{\sum_{i=1; p'_{ij} \neq 0}^m a_i x_i p'_{ij} c_i^w\}] \end{aligned}$$

representing similar failure costs for all undetected type i defects at stage j . Thus $U(F_m) + \sum_{j=m+1}^{n-1} L_m(j)$ provides a readily calculated lower bound on all solutions $F' = (f'_1, \dots, f'_{n-1})$ corresponding to the partial solution F_m (where $f_1 = f'_1, \dots$,

⁶ I wish to thank J. Heffernan for his comments in this area.

$f_m = f'_m$). These bounds may be improved through the inclusion of appraisal and detection costs $d_k(1)$, $j < k \leq n - 1$; however, measures must be taken to avoid possible double counting in the calculation of $\sum_{j=m+1}^{n-1} L_m(j)$.

3. An Illustrative Problem

For the simple eight-stage process of Figure 1, let the total number of units of production to be completed be 100 units per unit time. Assume further that the costs of appraisal and detection at each stage are independent, linear and of the form $d_i = m_i f_i$ for $i = 1, \dots, 7$.

Failure costs occur as repairs are initiated and completed. Assume that all direct costs may be classified as either "material" or "labor" costs. For the purpose of choosing between alternative screening programs, fixed costs common to all alternatives are assumed irrelevant. External failure costs are incurred as defects reach the warehouse and are subsequently shipped to the customer. Assume that all defects will be discovered by the customer and when found are repaired in the field. Let the direct cost of materials to repair a type i defect be α_i , the necessary labor required to effect the repair be some percent z_{is} of the value of the completed unit, and the per unit cost of ill will associated with a type i defect found by the customer be w_i . The direct per unit external failure cost of a type i defect is:

$$c_i^w = \alpha_i + z_{is}V_s + w_i \quad \text{for } i = 1, \dots, 7,$$

where V_s = the value of a completed unit of production.

Internal failure costs are those associated with the repair of defects detected at the various stages of production. Define the direct cost of materials to repair a type i defect as α_i regardless of when repair is undertaken. Let the labor required be a percent z_{ij} of the value of the unit when the repair is undertaken. If V_j represents the value of a unit of production at stage j , the direct per unit internal failure cost of a type i defect detected at a specific stage j is:

$$c_{ij}^r = \alpha_i + z_{ij}V_j \quad \text{for } i = 1, \dots, 7 \quad \text{and } j = 1, \dots, 7.$$

Finally for illustrative purposes, assume that a fixed cost of maintaining the internal repair facility as well as the external repair facility exists and is C_r and C_w respectively. These are accommodated in the analysis as follows. If $a_i x_i b_{is} > 0$ for any i , include C_w ; if $a_i x_i b_{io} > 0$ for any i include C_r . Let the cost parameters defined above be given as follows:

TABLE I
Production Stage/Type of Defect

	1	2	3	4	5	6	7	8	Dimension
a_i	0.20	0.10	0.10	0.10	0.10	0.10	0.05	—	
m_i	1,350	450	750	600	525	600	1,350	—	\$/per unit time
α_i	0.20	0.80	0.70	0.50	0.60	0.70	0.80	—	\$/unit
w_i	9.00	9.00	9.00	9.00	9.00	9.00	9.00	—	\$/unit
z_{ij}	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	% of V_j * for all j
v_j	10	25	28	43	83	116	249	249	\$/unit
$C_r = 200$ \$/t									
$C_w = 80$ \$/t									

Directly from the flow diagram of Figure 1, the connectivity matrix G is developed and shown in Figure 3. For any solution F , the elements of matrices P , P' and B are readily obtained using the equations developed for expected defect flow rates. Table II summarizes the branch and bound backtrack procedure employed to yield the optimal screening program $F^* = (0, 0, 0, 0, 1, 1, 0)$ of \$1,901.1.

TABLE II
Summary of Solution Procedure

Solution F_m or F	Lower Bound		Upper Bound
	$[U(F_m) + \sum_{j=m+1}^{n-1} L_m(j)]$ or $T(F)$		
(0, 0, 0, 0, 0, 0, 0)		\$2,734.3	\$2,734.3
(0, 0, 0, 0, 0, 0, 1)		2,539.3	2,539.3
(0, 0, 0, 0, 0, 1, 0)	\$1,494.1		2,539.3
(0, 0, 0, 0, 0, 1, 1)		2,484.9	2,484.9
(0, 0, 0, 0, 1, 0, 0)		2,899.9	2,484.9
(0, 0, 0, 0, 1, 0, 1)	1,595.7		2,484.9
(0, 0, 0, 0, 1, 1, 0)		2,350.5	2,350.5
(0, 0, 0, 0, 1, 1, 1)		2,765.5	2,350.5
(0, 0, 0, 1, 0, 0, 0)	1,645.9		2,350.5
(0, 0, 0, 1, 0, 0, 1)		1,901.1	1,901.1
(0, 0, 0, 1, 0, 1, 0)		3,126.1	1,901.1
(0, 0, 0, 1, 0, 1, 1)			1,901.1
(0, 0, 0, 1, 1, 0, 0)	2,001.5		1,901.1
(0, 0, 0, 1, 1, 0, 1)	1,913.5		1,901.1
(0, 0, 0, 1, 1, 1, 0)	1,710.5		1,901.1
(0, 0, 1, 0, 0, 0, 0)		2,979.9	1,901.1
(0, 0, 1, 0, 0, 0, 1)		2,854.9	1,901.1
(0, 0, 1, 0, 0, 1, 0)	1,809.7		1,901.1
(0, 0, 1, 0, 0, 1, 1)		2,530.5	1,901.1
(0, 0, 1, 1, 0, 0, 0)		3,215.5	1,901.1
(0, 0, 1, 1, 0, 0, 1)	2,048.1		1,901.1
(0, 0, 1, 1, 0, 1, 0)	2,255.3		1,901.1
(0, 0, 1, 1, 0, 1, 1)	2,167.3		1,901.1
(0, 0, 1, 1, 1, 0, 0)	1,860.9		1,901.1
(0, 0, 1, 1, 1, 0, 1)		2,630.7	1,901.1
(0, 0, 1, 1, 1, 1, 0)		3,315.7	1,901.1
(0, 1, 0, 0, 0, 0, 0)			1,901.1
(0, 1, 0, 0, 0, 0, 1)			1,901.1
(0, 1, 0, 0, 0, 1, 0)	2,530.9		1,901.1
(0, 1, 0, 0, 0, 1, 1)	2,494.5		1,901.1
(0, 1, 0, 1, 0, 0, 0)	2,647.9		1,901.1
(0, 1, 0, 1, 0, 0, 1)	2,670.3		1,901.1
(0, 1, 0, 1, 0, 1, 0)	2,299.3		1,901.1
(0, 1, 0, 1, 0, 1, 1)			1,901.1
(0, 1, 1, 0, 0, 0, 0)			1,901.1
(0, 1, 1, 0, 0, 0, 1)			1,901.1
(0, 1, 1, 0, 0, 1, 0)			1,901.1
(0, 1, 1, 0, 0, 1, 1)			1,901.1
(0, 1, 1, 1, 0, 0, 0)			1,901.1
(0, 1, 1, 1, 0, 0, 1)			1,901.1
(0, 1, 1, 1, 0, 1, 0)			1,901.1
(0, 1, 1, 1, 0, 1, 1)			1,901.1
(0, 1, 1, 1, 1, 0, 0)			1,901.1
(0, 1, 1, 1, 1, 0, 1)			1,901.1
(0, 1, 1, 1, 1, 1, 0)			1,901.1
(0, 1, 1, 1, 1, 1, 1)			1,901.1

Given $n = 8$, a total of 128 extreme point solutions exist with associated total expected costs $T(F)$ ranging from \$1,901.1 to \$6,012. The frequently employed program of 100 percent final inspection yields an expected cost of \$2,539; the decision to undertake no screening at all (the origin solution screening plan) results in a total expected cost of \$2,734. For this example, the cost is responsive to the screening program selected since the poorest program, $F = (1, 1, 1, 1, 1, 1, 1)$, generates a cost approximately 3.2 times that of the optimal program.

A sensitivity analysis indicates that the optimal screening program is quite stable. Varying parameters w_i , m_i , z_{ij} and α_i plus and minus fifty percent leaves the optimal program $F^* = (0, 0, 0, 0, 1, 1, 0)$ unchanged. Although a decrement of fifty percent

		TO PRODUCTION STAGE							
		1	2	3	4	5	6	7	8
FROM PRODUCTION STAGE	1	0	1	1	1	1	1	0	0
	2	0	0	0	0	1	0	0	0
	3	0	0	0	0	1	1	0	0
	4	0	0	0	0	0	1	0	0
	5	0	0	0	0	0	0	1	0
	6	0	0	0	0	0	0	1	0
	7	0	0	0	0	0	0	0	1
	8	0	0	0	0	0	0	0	0

FIGURE 3. The connectivity matrix G for the system of Figure 1.

in a_i causes the optimal program to shift to the origin solution $F = (0, 0, 0, 0, 0, 0, 0, 0)$ an increase of fifty percent in a_i leaves the optimal program unchanged.

4. Conclusions

The primary contribution of this and earlier models lies in the form of recommendations of courses of action for quality control screening. For nonserial production processes operating under quasi-concave cost structures, optimal screening programs remain *extreme point solutions*. Serial production systems may be viewed as a special case of this model, in which the connectivity matrix G is of the form $g_{ij} = 1$ if $j = i + 1$ and $g_{ij} = 0$ otherwise. Hence, for serial production systems under linear inspection cost structures, the extreme point solutions parallel and support the earlier findings of [9] and [11]. This has now been extended to nonserial production systems and a special class of nonlinear cost structures.

Optimal screening plans appear to be quite stable, and are influenced most by the proportions of defective materials added at each production stage. Consequently, a great deal of effort should be devoted to the definition and estimation of these parameters of input quality.

The model is first limited to those settings in which all defects are repairable and where the repaired units of production can be reinserted into the production process. Secondly, it is assumed that all defect types have an equal probability of being found at any given inspection station. By defining a decision variable f_{ij} as the probability of detecting a type i defect at station j , this latter constraint may be eliminated. The result is a set of matrices P , P' and B , one for each type of defect i . The methodology remains essentially the same.

The model does, however, accommodate per unit failure costs based upon the types and quantities of defects contained in each unit of production. This is often the case in practice; for example, the electrical industry is characterized by expensive and complex electronic assemblies which are primarily repaired through the replacement of *individual defects*.

The proposed matrix formulations provide an orderly and effective method of evaluating alternative screening plans. Technological process considerations (G),

target production outputs (N) and input quality characteristics (A) combine directly with relevant cost data (C and D) to yield the total expected cost of quality for any screening plan (F).

For many production processes, screening efforts strategically placed may present one of the most effective means of achieving desired quality levels at minimum total production costs. In failing to view screening as an acceptable means of quality control, producers are deciding (by default) in favor of the origin solution screening plan. As shown in the illustrative problem, nonobvious optimal screening programs may exist and provide substantial economies of production over both the origin solution plan and the prevalent 100 percent final inspection plan.

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